

NGC 6946-BH1 and the Vanishing-Star Problem in CPTG

A Finite Structural-Collapse Interpretation

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Abstract

Vanishing stars reverse the usual logic of transient astronomy. Instead of identifying stellar death through a luminous supernova, one searches for a massive star that was present, undergoes only a weak transient or no normal explosion, and then disappears. Motivated by the long-standing deficit of high-mass red-supergiant supernova progenitors, systematic failed-supernova searches now treat disappearance itself as an observable. NGC 6946-BH1 is one of the best-studied and strongest candidates: a luminous red-supergiant progenitor produced a weak 2009 transient, vanished optically, failed to recover a normal stellar photosphere, and remains associated with a red mid-infrared source. Recent JWST analysis separates four neighboring near-infrared stars from the remnant and gives $\log_{10}(L_{\text{rem}}/L_{\odot}) = 4.727^{+0.026}_{-0.025}$ and $\log_{10}(L_{\text{prog}}/L_{\odot}) = 5.632^{+0.023}_{-0.018}$ at 95% confidence, corresponding to $R_L = 0.1245$. Updated LBT monitoring through September 2025 remains compatible with no detectable optical recovery.

General relativity robustly permits black-hole formation, while failed-supernova phenomenology requires additional stellar-collapse physics and the classical singularity theorems lead to geodesic incompleteness rather than a derived finite interior. Curvature Polarization Transport Gravity (CPTG) retains the GR-controlled exterior spacetime and trapping horizon while allowing curvature polarization and transport to become dynamically significant deeper in the extreme compact regime. Its admissible compact solution class is finite-curvature and continuously matched to the GR exterior. The multiwavelength record of NGC 6946-BH1 is strongly compatible with that finite structural-collapse channel, making the source a direct observational target for the CPTG compact sector and placing vanishing stars within a broader population test of finite compact-state formation.

1 Vanishing Stars as an Observational Phenomenon

Most observational searches for stellar death are triggered by something that appears: a bright supernova, a gamma-ray burst, a neutrino burst, or a gravitational-wave event. A failed supernova presents the inverse problem. A massive star can undergo core collapse without producing a normal luminous supernova, so the decisive signal may be that a previously identified star simply disappears. Kochanek et al. proposed this “survey about nothing” strategy explicitly: monitor very large samples of nearby supergiants and determine whether each star ends its life with a conventional explosion or instead falls out of sight [1].

The motivation predates NGC 6946-BH1. Direct searches for core-collapse supernova progenitors revealed a deficit of the most luminous red supergiants, the so-called red-supergiant problem. Stars exist in the relevant high-mass red-supergiant range, yet they are underrepresented among securely identified progenitors of ordinary Type II supernovae. One long-standing interpretation is that some of these stars collapse to black holes without producing a canonical supernova [2]. The Large Binocular Telescope failed-supernova program was designed to test that possibility directly by monitoring luminous stars in nearby galaxies for disappearance rather than brightening [1, 3, 5].

That program found NGC 6946-BH1. Its progenitor was a luminous red supergiant that brightened in a relatively weak event in 2009 and then vanished from ordinary optical monitoring. Continued observations have not recovered the original star. Basinger et al. summarized the empirical situation unusually strongly: the progenitor was a massive luminous star and it subsequently vanished in the optical [4]. An eleven-year LBT analysis continued to recover NGC 6946-BH1 as the survey’s principal failed-supernova candidate and produced a broad estimate of the failed-supernova fraction, with large statistical uncertainty [5].

The observational class is now broader than one object. The VASCO project searches historical and modern sky surveys for sources that disappear between epochs [13]. M31-2014-DS1 provides a second massive-star disappearance event in which a luminous star brightened in the infrared and then faded without a normal supernova. De et al. explicitly interpret that event together with the earlier disappearance in NGC 6946 as evidence for failed supernovae forming stellar-mass black holes [10]. Careful searches must remove catalog artifacts, variability, obscuration, mergers, and other astrophysical channels. After that screening, stellar disappearance remains a real observational domain and quiet massive-star collapse is an empirical compact-object problem rather than a merely theoretical curiosity.

1.1 Why the phenomenon remains a compact-object problem

Black-hole formation is one of the robust consequences of classical GR. The unresolved problem is the observed path from a normal massive star to a weak transient, optical disappearance, dusty remnant, and hidden compact object: that phenomenology is not supplied by the Einstein equations alone. It requires a stellar-collapse engine and a radiative/hydrodynamic model. The explosion mechanism for massive stars is itself

nontrivial, and failed-supernova calculations invoke uncertain combinations of shock failure, neutrino mass-energy loss, pressure-wave propagation, envelope ejection, fallback, and obscuration [6, 7]. Different progenitor structures can therefore produce substantially different observable outcomes even when the final compact object is conventionally labeled a black hole.

There is a second, more fundamental distinction. Once a trapped region forms, classical GR does not provide a finite, curvature-bounded material endpoint of the kind used in CPTG. Under the hypotheses of the singularity theorems, trapped collapse is tied to geodesic incompleteness [12]. Thus the conventional framework can describe the exterior black-hole geometry and can model a failed supernova through additional stellar physics, but it does not derive a regular finite interior closure as the physical end state of collapse.

CPTG addresses precisely that missing finite-continuation possibility. In the parent framework, GR remains controlling for the exterior spacetime, the trapping horizon, orbital dynamics, and the ordinary relativistic strong-field regime. Curvature polarization and transport become dynamically significant only deeper in the extreme compact regime, where CPTG restricts admissible physical compact solutions to finite-curvature states. A star can therefore disappear behind a GR-controlled trapped exterior while the deeper CPTG continuation remains finite rather than ending in an infinite-curvature physical state. Because the compact response is structural and source dependent rather than set by one universal mass threshold, analogous events can recur across a population.

The purpose of this paper is therefore direct: determine whether the observed state of NGC 6946-BH1 is the kind of event that CPTG finite structural collapse is meant to describe. The answer rests on the observations themselves—disappearance, luminosity suppression, infrared remnant structure, weak ejecta, and X-ray quietness—rather than on any unstable stellar-mass estimate.

2 CPTG Structural-Collapse Framework

CPTG is a nonlinear geometric framework in which ordinary matter sources curvature, curvature develops a polarization response, and organized curvature can be redistributed through transport [14, 15]. In black-hole physics, the parent theory keeps GR controlling the exterior spacetime, the trapping horizon, orbital dynamics, and the ordinary relativistic strong-field regime. The compact CPTG response is restricted to the deeper high-gradient region in which the polarization–transport contribution becomes dynamically significant.

The retained compact invariant is

$$C_c = g_j g_j + \kappa_c^2 (\partial_i g_j)(\partial_i g_j), \quad (1)$$

with normalized coordinate

$$\chi = \frac{C_c}{a_\star^2}, \quad a_\star > 0, \quad (2)$$

where $a_\star$ is the structural acceleration scale of the compact object and κ_c is the compact-

sector gradient length. Neither quantity is imported from the galaxy or cosmological reductions.

The registered high-gradient polarization representative is

$$\mathcal{L}_{\text{pol,c}} = \beta a_\star^{2/3} C_c^{1/3} = \beta a_\star^{4/3} \chi^{1/3}. \quad (3)$$

When the gradient term dominates, its constitutive derivative with respect to the retained field gradient scales as $|\partial g|^{-1/3}$. The high-gradient response therefore softens rather than diverges. Compact transport is likewise required to remain bounded, so the physical CPTG compact solution class is restricted to finite-curvature states. A normal luminous supernova is not a condition of this compact admissibility. That is why a vanishing star is a natural CPTG target: the exterior electromagnetic display and the deeper compact-state completion are distinct parts of the collapse problem.

2.1 Finite Local Center

The local behavior required by a regular finite-center branch can be stated without assuming the unknown global compact profile. Near a regular center, write

$$\Phi(r) = \Phi_0 + \alpha_c r^2 + O(r^4). \quad (4)$$

In local Cartesian coordinates,

$$g_j = -2\alpha_c x_j + O(r^3), \quad (5)$$

so that

$$\partial_i g_j = -2\alpha_c \delta_{ij} + O(r^2). \quad (6)$$

The contracted gradient remains finite,

$$(\partial_i g_j)(\partial_i g_j) = 12\alpha_c^2 + O(r^2), \quad (7)$$

while $g_j g_j = O(r^2)$. Therefore, for finite α_c and κ_c ,

$$C_c(0) = 12\kappa_c^2 \alpha_c^2 < \infty, \quad (8)$$

and on the positive $a_\star$ branch,

$$\chi(0) = \frac{12\kappa_c^2 \alpha_c^2}{a_\star^2} < \infty. \quad (9)$$

This is a conditional local result: if the completed compact solution occupies a regular-center branch, the retained CPTG invariant is finite at the center. The globally matched strong-field equations are what select the actual interior. The local expansion nevertheless displays explicitly the finite behavior admitted by the CPTG compact solution class.

2.2 Finite closure versus the classical GR endpoint

The regular-center expansion exhibits a compact continuation qualitatively different from the geodesically incomplete classical continuation identified by the singularity theorems. The exterior can remain relativistic and black-hole-like, with the trapping horizon controlled by GR, while the deeper physical source region belongs to a finite-curvature CPTG branch rather than being required to terminate in divergent curvature.

This distinction matters directly for vanishing stars. In the conventional picture, “failed supernova” describes an astrophysical route to black-hole formation: the shock does not produce a normal explosion, some envelope material may be weakly ejected, fallback may occur, and the star disappears. CPTG retains that successful observational description but adds a different possibility for the hidden completion. The remnant can possess a finite structural compact interior continuously matched to the GR-controlled exterior.

The mass controversy surrounding NGC 6946-BH1 is therefore secondary to the question addressed here. A changing estimate of progenitor, helium-core, terminal, or remnant mass does not change the defining observation that a massive star underwent a weak event and then vanished. Because the CPTG compact response is structural rather than mass-only, the candidacy of NGC 6946-BH1 does not depend on selecting one of those evolving mass estimates.

3 Observed State of NGC 6946-BH1

NGC 6946-BH1 is unusually valuable because its candidacy is built from a chronological, multiwavelength record. The source has a directly observed progenitor, a recorded weak transient, long-baseline optical disappearance, repeated infrared follow-up, a resolved JWST environment, a modeled dusty remnant, and an X-ray upper limit [3, 4, 8, 9, 11]. The case therefore rests on an observed sequence of state changes rather than on a single inferred stellar parameter. The updated LBT record now extends through September 2025: seven measurements added from October 2023 through September 2025 remain compatible with no detectable optical emission, and the post-outburst R -band light curve has a standard deviation below $10^3 L_\odot$ with no significant long-term trend [11].

Table 1: Observed and literature-derived quantities defining the vanishing-star state of NGC 6946-BH1 and its structural-collapse interpretation.

Quantity	Adopted value	Status and role in this paper
Object	NGC 6946-BH1 / N6946-BH1	Observed identification; vanishing-star and failed-supernova candidate.
Progenitor state	Luminous red supergiant	Direct pre-event stellar source; establishes the vanished progenitor.
Weak 2009 transient	Peak luminosity above $10^6 L_\odot$	Observed low-luminosity outburst context.
Remnant luminosity	$\log_{10}(L_{\text{rem}}/L_\odot) = 4.727^{+0.026}_{-0.025}$ at 95% confidence	JWST SED-model result; sets the late remnant luminosity scale.

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Quantity	Adopted value	Status and role in this paper
Progenitor luminosity	$\log_{10}(L_{\text{prog}}/L_{\odot}) = 5.632^{+0.023}_{-0.018}$ at 95% confidence	Revised progenitor SED fit; normalization for the luminosity-suppression test.
Luminosity ratio	$R_L = 0.1245$	Derived from the central SED values; failed-SN / survivor diagnostic used below.
Mid-infrared state	Red MIRI source; silicate dust shell	Observed/modelled obscured-remnant environment.
Short-wavelength remnant emission	Negligible below roughly $2\mu\text{m}$	JWST-model result; no exposed normal stellar SED is required.
Optical recovery	No detectable recovery through September 2025; post-outburst R -band scatter $< 10^3 L_{\odot}$	Long-baseline observational upper-limit state; survivor stress test.
X-ray counterpart	No Chandra detection	Observational upper limit; quiet-remnant constraint.
Dust/ejecta lower scale	$v_e \simeq 60 \text{ km s}^{-1}$; $M_e \gtrsim 0.08 M_{\odot}$; $K_e \gtrsim 3 \times 10^{45} \text{ erg}$	Model-derived lower limits; low-energy ejecta diagnostic.

The observed evolution can be summarized as

$$\begin{aligned}
&\text{weak transient} \downarrow \\
&\text{optical disappearance} \downarrow \\
&\text{red mid-infrared remnant} \downarrow \\
&\text{X-ray nondetection}
\end{aligned} \tag{10}$$

The importance of Eq. (10) is its coherence. Collectively, the stages support the quiet-collapse interpretation and reproduce the observational signature that failed-supernova surveys were designed to find: a massive star that ends its visible life without an ordinary supernova and without returning as a normal luminous star.

4 Luminosity Suppression and Infrared Remnant

The most direct scalar measure of the late-time suppression is the remnant-to-progenitor luminosity ratio,

$$R_L = \frac{L_{\text{rem}}}{L_{\text{prog}}}. \tag{11}$$

Using the central JWST reanalysis values [11],

$$\log_{10}(L_{\text{rem}}/L_{\odot}) = 4.727, \tag{12}$$

and

$$\log_{10}(L_{\text{prog}}/L_{\odot}) = 5.632, \tag{13}$$

gives

$$\log_{10} R_L = -0.905, \tag{14}$$

and therefore

$$R_L = 0.1245. \quad (15)$$

The remnant is therefore only about one eighth as luminous as the progenitor on the central adopted branch. That late-time suppression is qualitatively different from the merger-remnant population considered in the same JWST analysis: For’es-Toribio and Kochanek find merger remnants typically 10–100 times more luminous than their progenitors at comparable late phases, whereas the two failed-supernova candidates are approximately ten times dimmer than their progenitors [11]. Combined with the continued optical non-recovery and the red mid-infrared SED, Eq. (15) strengthens the failed-supernova interpretation independently of any CPTG mass decomposition.

The luminosity ratio is therefore one of the strongest pieces of the case. The late source is not recovering toward the luminosity of the vanished red supergiant; it remains deeply suppressed and embedded in a red dust-dominated environment. That is the direction expected for a genuinely hidden compact remnant, not an ordinary surviving star.

5 Dust-Ejecta Scale and X-ray Nondetection

The infrared remnant also provides a dynamical lower scale. The recent JWST dust-shell analysis associates the favored inner shell radius with an ejecta speed near

$$v_e \simeq 60 \text{ km s}^{-1}, \quad (16)$$

over the post-transient baseline and obtains lower limits

$$M_e \gtrsim 0.08 M_\odot, \quad (17)$$

and

$$K_e \gtrsim 3 \times 10^{45} \text{ erg} \quad (18)$$

for the favored silicate model [11]. These are model-dependent lower limits rather than a measurement of the complete explosion energy. They nevertheless establish a low-energy ejecta component consistent with the broader failed-supernova phenomenology.

The Chandra nondetection supplies a separate constraint. Basinger et al. report a 90% upper limit of approximately $1500 L_\odot$ for a power-law spectral assumption. Their published normalization to a $10 M_\odot$ black hole gives

$$\frac{L_X}{L_{\text{Edd}}(10 M_\odot)} \lesssim 4.6 \times 10^{-3}. \quad (19)$$

The $10 M_\odot$ denominator is retained only because it is the published reference normalization used for the X-ray limit. It is not adopted here as the mass of NGC 6946-BH1 and plays no role in the structural-collapse classification.

The X-ray nondetection is consistent with the same quiet-remnant picture. Weak accretion, low radiative efficiency, and obscuration can each suppress the X-ray flux, so the nondetection is not a unique accretion diagnostic. Its force is cumulative: the source that vanished optically and remains infrared dominated also lacks a bright X-ray counterpart.

6 The Multiwavelength Case for Structural Collapse

The structural-collapse case is strongest when the observations are treated together rather than as isolated diagnostics. NGC 6946-BH1 satisfies five mutually reinforcing features of a quiet-collapse candidate:

1. a luminous massive progenitor was observed before the event;
2. the 2009 event was weak compared with a normal luminous core-collapse supernova;
3. the progenitor subsequently disappeared from ordinary optical monitoring and has not recovered;
4. the late source is red, dust dominated, and only $R_L = 0.1245$ as luminous as the progenitor on the current central JWST branch;
5. the inferred ejecta scale is low and no Chandra X-ray counterpart has been detected.

No single item proves the nature of the remnant. The combined sequence is what makes NGC 6946-BH1 exceptional. A dust-obscured survivor must explain why the bolometric remnant remains so strongly suppressed. A merger remnant must explain why the late source is far dimmer rather than substantially brighter than the progenitor population used for comparison by Forés-Toribio and Kochanek [11]. An ordinary supernova must explain the absence of an ordinary luminous explosion. Quiet compact-object formation naturally organizes all five observations under one event.

The shifting mass literature does not weaken this conclusion. Published values refer to different coordinates—initial mass, helium-core mass, terminal stellar mass, or modeled remnant mass—and move as stellar models and progenitor luminosities change. None of those choices alters the observed disappearance sequence. For the present question, mass is therefore supporting stellar context rather than the classifier.

7 What CPTG Adds to the Conventional Failed-Supernova Picture

A conventional failed-supernova model already captures an important part of the phenomenology: a massive stellar core can fail to launch a normal explosion and form a black hole while neutrino mass loss or a weak outgoing pulse ejects part of the envelope [6, 7]. That is a serious astrophysical explanation for NGC 6946-BH1, not a straw man. The unresolved issue is that the conventional picture is assembled from two physically distinct

layers: stellar-collapse microphysics determines whether the star explodes, while classical GR supplies the compact spacetime after trapped collapse.

Those layers leave two openings that are directly relevant to CPTG. First, failed-supernova phenomenology is highly model dependent. The strength of the transient, the amount of ejected envelope, fallback, and the surviving dust distribution depend on progenitor structure and uncertain collapse physics. Second, the classical singularity theorems do not furnish a finite structural interior completion. Under their standard hypotheses, trapped collapse implies geodesic incompleteness [12].

CPTG changes the second layer. In the extreme compact regime, curvature polarization and transport can become dynamically significant while GR continues to control the exterior spacetime and trapping horizon. CPTG restricts the admissible deeper compact solution to finite-curvature states. A normal luminous explosion is not a requirement of that compact solution class, so a quiet disappearance is compatible with the same finite-collapse interpretation rather than being an observational afterthought.

Observed or physical issue	Conventional failed-SN / GR picture	CPTG structural-collapse picture
Weak or absent normal supernova	Explosion failure plus model-dependent neutrino-driven or hydrodynamic envelope response	The CPTG compact solution class does not require a luminous explosion; weak external response is compatible
Optical disappearance	Compact black-hole formation is a viable interpretation	Compatible with formation of a trapped compact state having a finite deeper continuation
Weak ejecta and dusty remnant	Produced by envelope response, fallback, recombination, and dust; outcome depends on progenitor model	Compatible aftermath of a finite compact remnant with limited external ejection
Hidden compact interior	Classical exterior is well defined; trapped collapse is associated with geodesic incompleteness under the usual theorem hypotheses	Finite-curvature compact interior is the admissible CPTG completion; a regular-center branch gives the local finite form used here
Population of disappearing stars	Failed-SN fraction depends on stellar explosability and progenitor demographics	A recurrent population is expected qualitatively if massive stars repeatedly enter the CPTG high-gradient compact regime

Table 2: GR readily permits black-hole formation. The comparison concerns the hidden compact continuation and the physical interpretation of quiet disappearance: a model-dependent failed explosion with a GR-controlled exterior, or the same exterior joined to a finite-curvature CPTG compact interior.

For NGC 6946-BH1, both pictures can accommodate the observed disappearance at the level of external phenomenology. CPTG becomes scientifically relevant because it offers a finite compact interpretation that classical GR does not supply and because the same mechanism naturally extends to other vanishing-star candidates.

8 Vanishing Stars as a Population Test of Structural Collapse

The strongest implication of the CPTG interpretation is therefore not an exact mass for NGC 6946-BH1. It is a population statement. If the finite high-gradient CPTG compact branch is a genuine compact-object channel, NGC 6946-BH1 should not be unique. Other

massive stars that enter the extreme compact regime in which the polarization–transport response becomes dynamically significant should also be capable of disappearing with weak or highly suppressed electromagnetic explosions.

The observational program needed to test that statement already exists. The LBT survey monitors luminous stars specifically to find failed supernovae, VASCO searches for disappearing sources across long historical baselines, and M31-2014-DS1 supplies an additional massive-star disappearance case [5, 13, 10]. De et al. explicitly interpret M31-2014-DS1 together with the earlier NGC 6946 event as evidence for failed supernovae forming stellar-mass black holes [10]. The eleven-year LBT analysis inferred a failed-supernova fraction $f = 0.16^{+0.23}_{-0.12}$ at 90% confidence under the assumption that NGC 6946-BH1 was the one clear failed-SN detection [5]. The interval is broad, but the survey is directly constraining a potentially non-zero failed-supernova population channel, with a central estimate already in the theoretically interesting range.

The population test is specific: after contaminants and ordinary stellar explanations are removed, a population of genuine quiet-collapse remnants should remain if the finite CPTG compact branch is a real compact-object channel. Their frequency, progenitor dependence, remnant luminosities, dust evolution, and eventual gravitational signatures can then test whether the same high-gradient compact response recurs across stellar collapse events.

9 Conclusion

NGC 6946-BH1 is not mysterious because a massive star is forbidden to form a black hole. It is mysterious because the star was observed before collapse, produced no normal luminous supernova, vanished optically, and then persisted only as a much dimmer red infrared remnant. The event gives astronomy a rare direct view of a stellar death channel that ordinary transient surveys are designed to miss.

The evidence for quiet collapse is unusually coherent. The progenitor was a luminous red supergiant. The 2009 transient was weak. The star did not return. JWST now separates neighboring near-infrared stars from the mid-infrared candidate remnant and gives

$$R_L = 0.1245, \tag{20}$$

so the late source is only about one eighth as luminous as the progenitor on the central current branch. The remnant is dust dominated, the ejecta scale is low, and no Chandra counterpart has been detected. Taken together, these are the defining observations of a strong vanishing-star / failed-supernova candidate.

The conventional interpretation is black-hole formation after a failed or weak explosion. That explanation remains viable, but it leaves the collapse phenomenology dependent on uncertain stellar microphysics, while the classical singularity-theorem description of trapped collapse terminates in geodesic incompleteness rather than a derived finite interior. CPTG supplies an additional physical possibility: the extreme compact response selects a finite-curvature trapped solution continuously matched to the GR-controlled exterior.

That interpretation fits the phenomenon naturally. A luminous explosion is not a requirement of the CPTG compact solution class, and the compact response is not determined by one stellar mass estimate. Quiet disappearance can therefore recur if massive stars repeatedly enter the relevant high-gradient compact regime. The existence of systematic failed-supernova surveys, additional disappearance candidates, and long-baseline vanishing-source searches makes this a population-level question rather than a one-object curiosity.

The result of this paper is therefore affirmative and limited in exactly the useful way: *NGC 6946-BH1 is a strong observational candidate for CPTG finite structural collapse.* The significance of that conclusion is the existence of a real astrophysical event class for which CPTG supplies a finite compact-state explanation absent from the standard classical interior picture. Future vanishing-star statistics and direct strong-field remnant observations can determine whether that possibility is merely allowed or is realized in nature.

Data, Code, and Evidence Availability

The observational quantities used in this paper are taken from the cited optical, infrared, failed-supernova monitoring, vanishing-source, and X-ray studies. No compact-remnant mass estimate is fitted, selected, or converted in the present analysis. The numerical luminosity ratio is derived directly from the cited progenitor and remnant luminosities. CPTG theory papers, supporting calculations, and public research resources are maintained through the project repository [16].

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